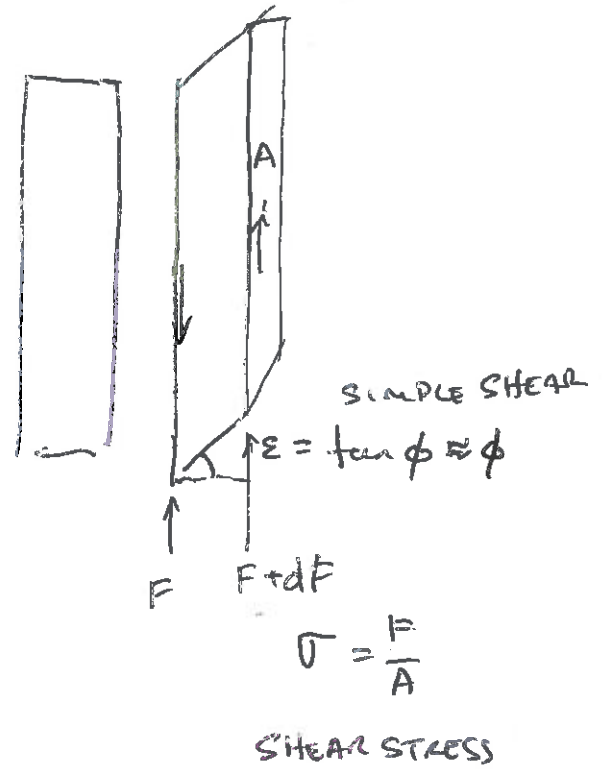
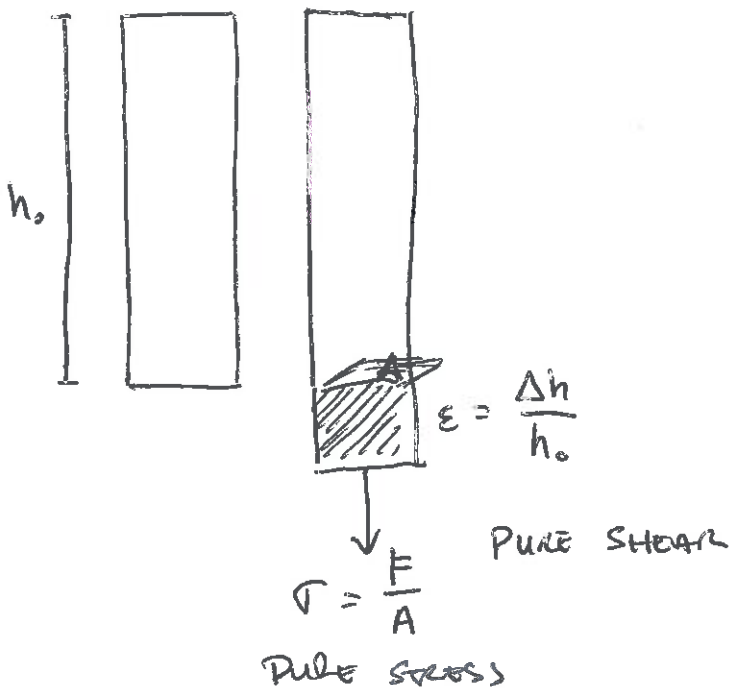
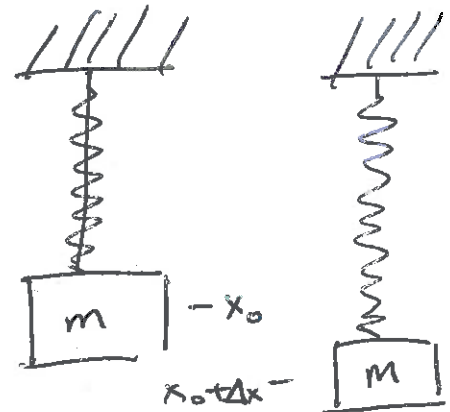
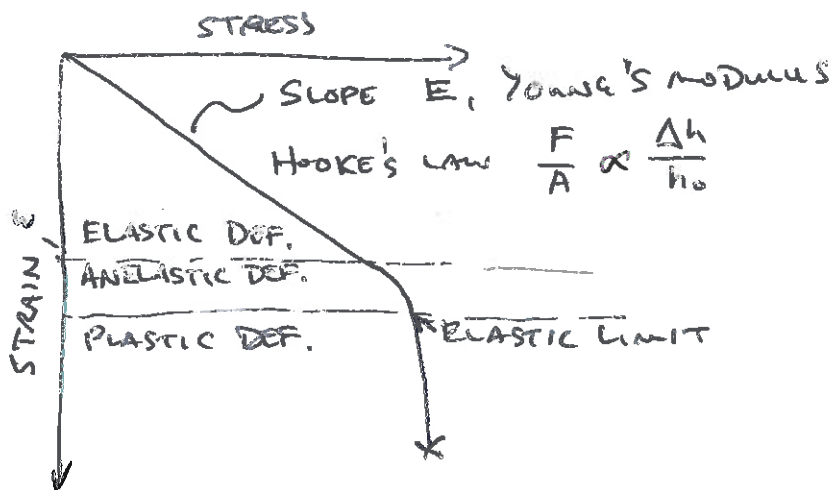


STRESS STRAIN



$$\tau = \frac{F}{A} \propto \frac{\Delta h}{h_0} = \epsilon$$

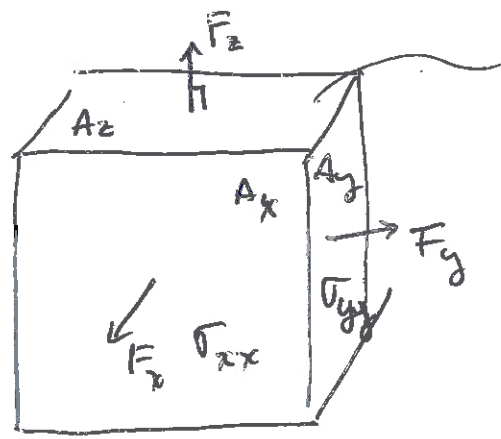
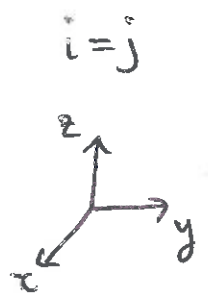


$F = k \Delta x$ HOOKE'S LAW

RESTORING FORCE

SPRING CONSTANT

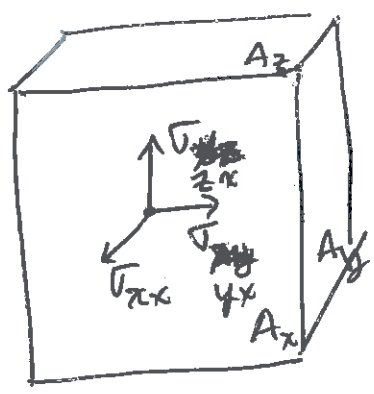
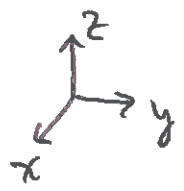
$$\sigma_{ij} = \lim_{A_j \rightarrow 0} \frac{F_i}{A_j}$$



$$\sigma_{zz} = \lim_{A_z \rightarrow 0} \frac{F_z}{A_z}$$

NORMAL STRESS
"PURE"

$i \neq j$



$$\sigma_{ij} = \lim_{A_j \rightarrow 0} \frac{F_i}{A_j}$$

SHAPE
SHEAR STRESS

u - DISPLACEMENT FIELD

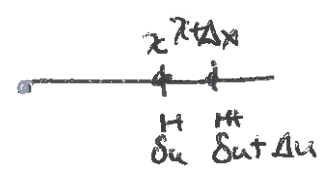
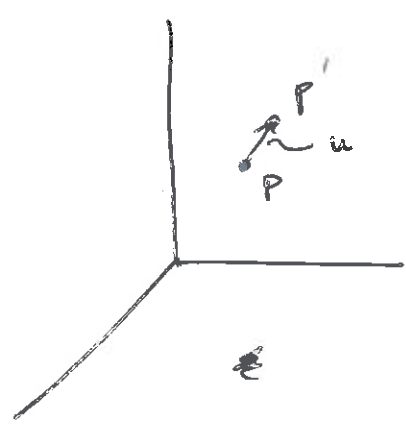
$$u = u(x, y, z)$$

NORMAL STRAIN

$$\epsilon_{xx} = \frac{\partial u_x}{\partial x}$$

$$\epsilon_{yy} = \frac{\partial u_y}{\partial y}$$

$$\epsilon_{zz} = \frac{\partial u_z}{\partial z}$$



$$\frac{\Delta V}{V} = \Delta = \epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz}$$

DILATION

SCALAR DISPLACEMENT

POTENTIAL

$$\Delta = \nabla \cdot u$$

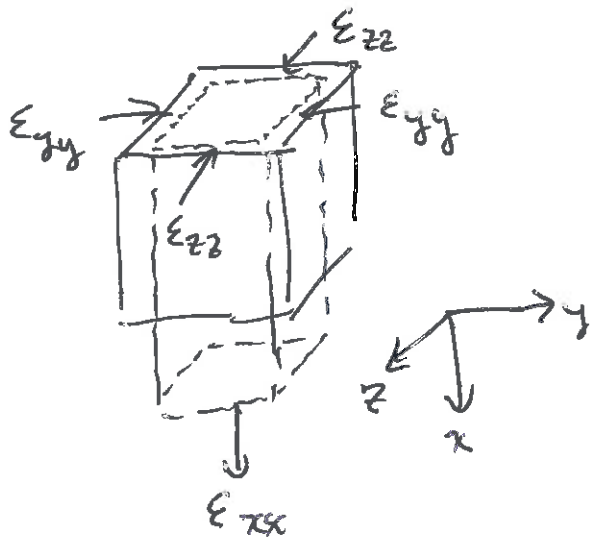
$$\frac{\partial^2 \Delta}{\partial t^2} = \alpha^2 \nabla^2 \Delta$$

↑

WAVE VELOCITY

P-WAVE

$$V_p = \alpha = \sqrt{\frac{K + \frac{4}{3}\mu}{\rho}}$$



K - BULK MODULUS [P_a]

μ - SHEAR MODULUS [P_a]

ρ - DENSITY (kg m⁻³)

$$\epsilon_{xx} = -\nu \epsilon_{yy}$$

$$\epsilon_{xx} = -\nu \epsilon_{zz}$$

ISOTROPIC

SAME PROPERTIES IN ALL DIMENSIONS

ν - POISSON'S RATIO

$$\sigma_{xx} = E \epsilon_{xx}$$

E = YOUNG'S MODULUS

~~$$E = \frac{3K(1-2\nu)}{1+\nu}$$~~

$$E = 3K(1-2\nu)$$

$$E = 2\mu(1+\nu)$$

SHEAR WAVES

$$\frac{\partial^2 \vec{\psi}}{\partial t^2} = \beta^2 \nabla^2 \vec{\psi}$$

$$V_s = \beta = \sqrt{\frac{\mu}{\rho}} \quad \text{SHEAR VELOCITY}$$